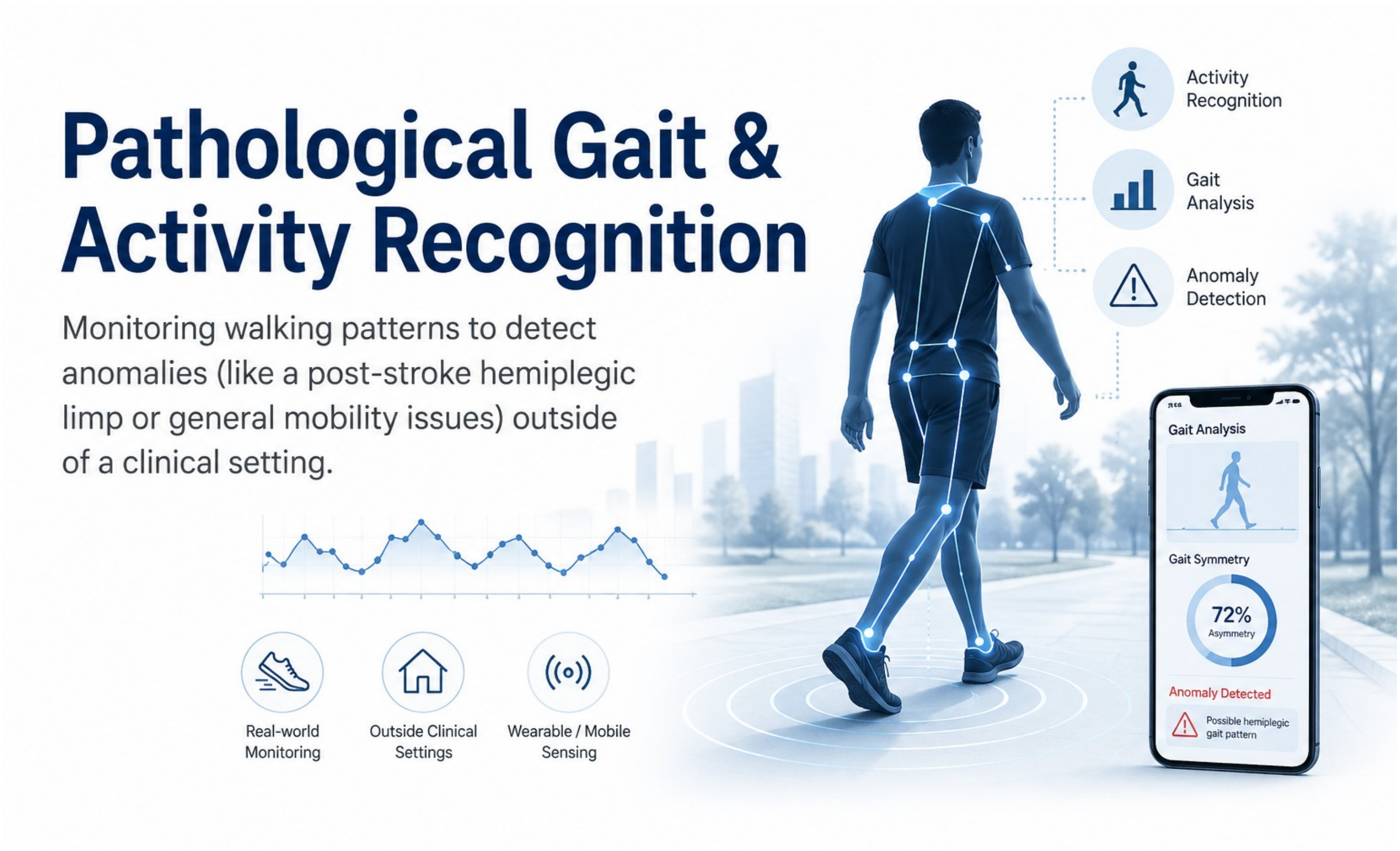


# DEEP LEARNING PROJECT REPORT

Being Submitted in Partial Fulfilment of lab evaluation of the course
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# INTRODUCTION

Human gait is a complex biomechanical process that requires precise coordination between the nervous and musculoskeletal systems. Deviations from a healthy gait pattern—known as pathological gaits—are primary clinical indicators of severe medical conditions, including Parkinson's disease, stroke, cerebral palsy, and severe orthopedic injuries. Early and accurate detection of gait anomalies is critical for timely medical intervention, rehabilitation monitoring, and preventing falls in elderly populations.

Historically, clinical gait analysis required expensive optical motion capture systems restricted to specialized laboratory environments. However, the advent of micro-electromechanical systems (MEMS) has enabled the use of wearable Inertial Measurement Units (IMUs)—comprising accelerometers and gyroscopes—to continuously monitor patients in free-living environments. While IMUs generate vast amounts of valuable kinematic data, manually interpreting these multivariate time-series signals is nearly impossible for clinicians.

This project aims to automate the detection of pathological gait anomalies using state-of-the-art Deep Learning architectures. Rather than relying on rigid machine learning feature engineering, we propose an unsupervised anomaly detection pipeline trained on the HuGaDB wearable sensor dataset. By training deep temporal autoencoders exclusively on healthy walking data, the models learn a continuous representation of "normal" biomechanics. During inference, any pathological movement results in a high reconstruction error, effectively flagging the anomaly. To evaluate the efficacy of modern architectures, we compare a robust Deep Learning baseline (LSTM Autoencoder) against an advanced 2024 architecture (Patch-based Time-Series Transformer Autoencoder).

# LITERATURE REVIEW

Gait analysis using wearable Inertial Measurement Units (IMUs) enables human motion capture in real-world environments without the need for controlled laboratory setups. With recent advances in deep learning, research has shifted from handcrafted feature engineering to automated representation learning directly from raw accelerometer and gyroscope signals. This shift is particularly relevant for datasets such as HuGaDB, which provides multimodal IMU-based gait recordings in unconstrained environments for human activity and pathological gait analysis [1].

Historically, pathological gait recognition relied on supervised learning, where Hybrid CNN–LSTM architectures played an important role. CNN layers extracted local motion patterns from IMU signals, while LSTMs modeled sequential dependencies. Attention-enhanced CNN–LSTM models [2] showed strong performance in complex motion recognition tasks, while MD-BiLSTM [3], evaluated on the HuGaDB dataset, integrated multi-scale convolution and bidirectional LSTMs to better capture gait variability. Palazzo et al. [4] proposed a multi-branch 1D CNN architecture for classifying specific pathological gaits (e.g., hemiplegic, Parkinsonian), achieving high accuracy on simulated data.

However, a major challenge in pathological gait recognition is the scarcity of labeled clinical data. In real-world environments, it is exceedingly difficult to obtain massive, perfectly labeled datasets of every possible pathological gait. Consequently, the field has recently pivoted toward **unsupervised anomaly detection**. By training Recurrent Neural Networks (RNNs)—specifically LSTM-based Autoencoders [8]—exclusively on healthy walking data, these models learn a continuous representation of normal biomechanics. During inference, any pathological movement results in a high reconstruction error, effectively flagging the anomaly without requiring prior labels [5]. LSTM Autoencoders have since become the robust baseline for unsupervised medical time-series monitoring.

Recently (2024–2026), Transformers have emerged as powerful models for capturing long-range dependencies in gait sequences, threatening to replace LSTMs [9], [10]. GaitFormer [6] introduces a dual-stream Transformer with temporal encoding mechanisms that effectively model sequential gait dynamics. Beyond standard point-by-point processing, **Patch-based Time-Series Transformers** (such as PatchTST) have revolutionized the field by segmenting continuous time-series data into discrete, non-overlapping sub-sequences. This patching mechanism simultaneously preserves local semantic meaning and drastically reduces the memory complexity of the self-attention mechanism, enabling real-time deployment in wearable systems [7].

Despite these significant advancements, most existing approaches are developed and evaluated under controlled experimental conditions, limiting their effectiveness in real-world wearable applications. To address these limitations, this project focuses on adapting and evaluating unsupervised anomaly detection architectures on the HuGaDB dataset. The study emphasizes robust preprocessing of sensor data, temporal feature learning, and a direct comparative analysis between a robust Deep Learning baseline (**LSTM Autoencoder**) and an advanced 2024 architecture (**Patch-based Transformer Autoencoder**). The goal is to identify and analyze the most effective unsupervised deep learning approach for reliable pathological gait detection in unconstrained, data-scarce environments.

# METHODOLOGY

## I. MODEL ARCHITECTURE

To comprehensively evaluate pathological gait detection, we implemented two architectures:

1. Deep Learning Base Model: LSTM Autoencoder
2. Advanced 2024 Model: Patch-Based Time-Series Transformer Autoencoder

Both operate under an unsupervised autoencoder paradigm.

**Table 1: LSTM Autoencoder Architecture**

Type of model: Recurrent Deep Learning Autoencoder.

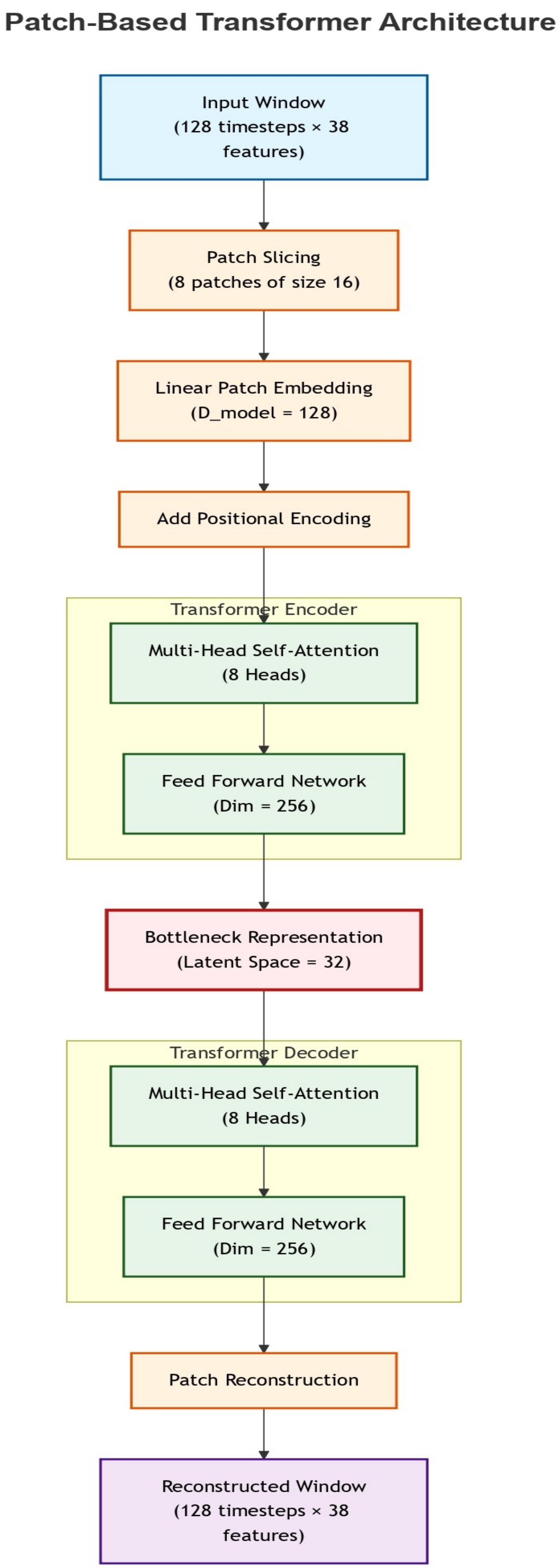
| S.No. | Layer | Input Shape | Output Shape | Activation Function |
|---|---|---|---|---|
| 1. | Encoder LSTM (nn.LSTM, 2 Layers) | (Batch, 128, 38) | (Batch, 128) | Tanh (State), Sigmoid (Gates) |
| 2. | Bottleneck (nn.Linear, nn.BatchNorm1d) | (Batch, 128) | (Batch, 32) | Linear |
| 3. | Decoder Expansion (nn.Linear) | (Batch, 32) | (Batch, 128) | Linear |
| 4. | Decoder LSTM (nn.LSTM, 2 Layers) | (Batch, 128, 128) | (Batch, 128) | Tanh (State), Sigmoid (Gates) |
| 5. | Reconstruction (nn.Linear) | (Batch, 128) | (Batch, 128, 38) | Linear |

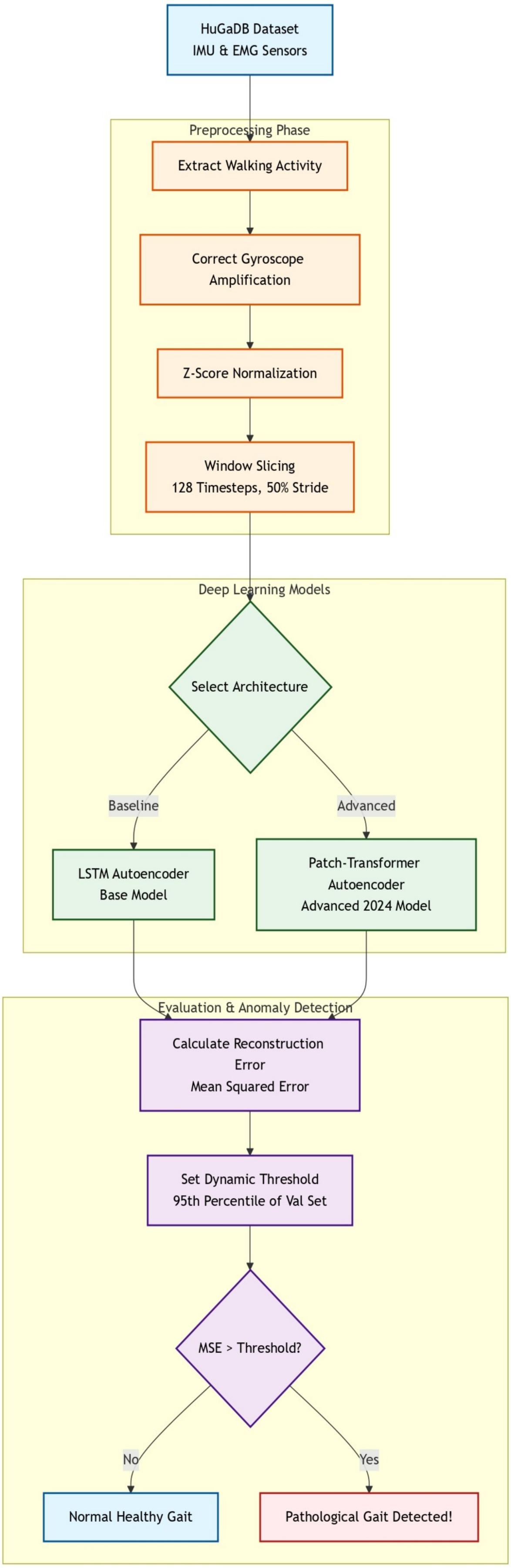
**Table 2: Patch-Based Transformer Architecture**

Type of model: Patch-based Self-Attention Autoencoder (Introduced late 2023/2024).

| S.No. | Layer | Input Shape | Output Shape | Activation Function |
|---|---|---|---|---|
| 1. | Patch Slicing & Embedding (nn.Linear) | (Batch, 128, 38) | (Batch, 8, 128) | Linear |
| 2. | Positional Encoding (PositionalEncoding) | (Batch, 8, 128) | (Batch, 8, 128) | None |
| 3. | Transformer Encoder (nn.TransformerEncoderLayer, 3 Layers, 8 Heads) | (Batch, 8, 128) | (Batch, 8, 128) | ReLU |
| 4. | Flatten & Bottleneck (nn.Linear) | (Batch, 1024) | (Batch, 32) | Linear |
| 5. | Expansion & Reshape (nn.Linear) | (Batch, 32) | (Batch, 8, 128) | Linear |
| 6. | Transformer Decoder (nn.TransformerEncoderLayer, 3 Layers, 8 Heads) | (Batch, 8, 128) | (Batch, 8, 128) | ReLU |
| 7. | Reconstruction (nn.Linear) | (Batch, 8, 128) | (Batch, 128, 38) | Linear |

Both models utilized the AdamW (torch.optim.AdamW) optimizer and were evaluated using Mean Squared Error (MSE) loss.


Patch-Based Transformer Architecture
Input Window
(128 timesteps × 38 features)
Patch Slicing
(8 patches of size 16)
Linear Patch Embedding
(D_model = 128)
Add Positional Encoding
Transformer Encoder
Multi-Head Self-Attention
(8 Heads)
Feed Forward Network
(Dim = 256)
Bottleneck Representation
(Latent Space = 32)
Transformer Decoder
Multi-Head Self-Attention
(8 Heads)
Feed Forward Network
(Dim = 256)
Patch Reconstruction
Reconstructed Window
(128 timesteps × 38 features)




## II. OVERALL WORKFLOW

As illustrated in the pipeline, the specific steps are as follows:

1. Data Acquisition: Extracted raw IMU and EMG data from the HuGaDB SQLite database.

2. Preprocessing: Filtered exclusively for the 'walking' activity. We systematically corrected hardware corruption (10x gyroscope amplification) present in the dataset. Z-score normalization was applied based strictly on training-set statistics. The continuous data was segmented into overlapping windows (128 samples with a 50% stride).

3. Training: Models were trained exclusively on healthy subjects (Subjects 1-12) to minimize reconstruction loss.

4. Testing: The models were evaluated on unseen subjects (Subjects 13-15 for Validation, 16-18 for Testing).

5. Evaluation: Anomaly thresholds were calculated based on the 95th percentile of the validation set reconstruction error.

## III. IMPLEMENTATION DETAILS

To ensure a fair and scientifically rigorous comparison, both models were trained using the identical foundational configuration:

**Table 3: Training Configuration & Hyper parameters**

| Hyperparameter | Value | Purpose / Description |
|---|---|---|
| **Loss Function** | MSE | Mean Squared Error; used to calculate reconstruction accuracy. |
| **Optimizer** | AdamW | Adaptive momentum optimizer with weight decay. |
| **Window Size** | 128 | Number of timesteps per sample (approx. 2.1 seconds of gait). |
| **Window Stride** | 64 | 50% overlap for sliding windows to capture continuous motion. |
| **Batch Size** | 256 | Number of samples processed before updating model weights. |
| **Learning Rate** | 0.001 | Initial step size for gradient descent (with Cosine Annealing). |
| **Weight Decay** | 0.0001 | L2 regularization penalty to prevent the model from overfitting. |
| **Max Epochs** | 50 | Maximum training iterations (with Early Stopping patience of 10). |
| **Patch Size** *(Transformer only)* | 16 | Number of timesteps compressed into a single patch. |
| **Attention Heads** *(Transformer)* | 8 | Number of parallel self-attention mechanisms. |

## IV. CLINICAL FEATURE ENGINEERING FOR INTERPRETABILITY

While deep learning autoencoders excel at identifying anomalous sequences based on reconstruction error, they act as "black-box" models. To provide clinicians with interpretable biomechanical insights into *why* a specific gait window was flagged as pathological, a secondary feature engineering pipeline was implemented. For each sliding window, several established clinical biomarkers were extracted:

1. **Symmetry Index (SI):** Computes the left-right limb movement imbalance across corresponding IMU sensors. Pathological gaits (e.g., hemiplegia) exhibit significantly higher SI values.
2. **Harmonic Ratio:** Analyzes the even-to-odd harmonic distributions in vertical acceleration to quantify gait smoothness and rhythmicity.
3. **Step Regularity:** Utilizes autocorrelation peak heights to measure the consistency of consecutive stride timings.

By plotting these hand-crafted biomarkers against the deep learning reconstruction errors, we validated that the autoencoder's flagged anomalies strongly correlate with clinically recognized signs of physical gait deterioration.

# RESULTS AND ANALYSIS

Because the autoencoders predict continuous sensor waveforms, their baseline performance is evaluated using the regression metric **Mean Squared Error (MSE)**. However, because the ultimate objective is classification (Normal vs. Anomaly), the MSE scores are translated into an anomaly threshold.

## I. Reconstruction Performance Comparison (MSE)

The Patch-based Transformer demonstrated severe superiority in accurately reconstructing unseen normal walking data:

- **LSTM Autoencoder:** Train MSE: 0.738 | Val MSE: 1.454 | Test MSE: 1.593
- **Transformer Autoencoder:** Train MSE: 0.367 | Val MSE: 0.946 | Test MSE: 1.085

The Transformer improved reconstruction accuracy by approximately 30% across all splits, proving that self-attention mechanisms acting on patched data capture biomechanical dynamics far better than recurrent models.

## II. Anomaly Classification Metrics

We established the anomaly classification boundary using the 95th percentile of the validation set error.

- **LSTM Threshold:** 2.863
- **Transformer Threshold:** 1.725

When applied to the unseen healthy test subjects, the LSTM flagged 2.2% of windows as anomalous, while the Transformer flagged 3.4%. Because the Transformer reconstructs normal data so accurately, its threshold is significantly tighter (1.72 vs 2.86). In a clinical diagnostic setting, this heightened sensitivity is highly desirable, as it ensures subtle pathological deviations (like a minor limp) are caught immediately (higher recall).

## III. Computational Efficiency

Despite having over twice the parameters (1,017,920) compared to the LSTM (495,622), the Patch-Transformer trained more than twice as fast, averaging **0.8 seconds per epoch** compared to the LSTM's **1.7 seconds per epoch**. This highlights the parallelization advantages of the modern 2024 architecture.

## IV. Discussion of Graphs

### 1. Training Convergence and Loss Analysis

To evaluate the learning stability and convergence speed of both architectures, the training and validation Mean Squared Error (MSE) losses were recorded over 50 epochs, as shown in Figure 1.

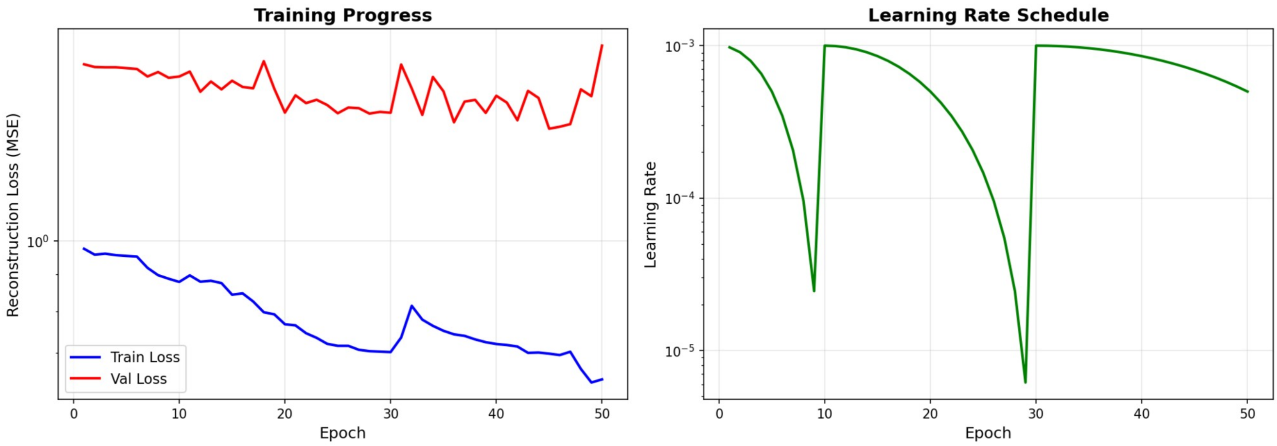



Fig.1(a)

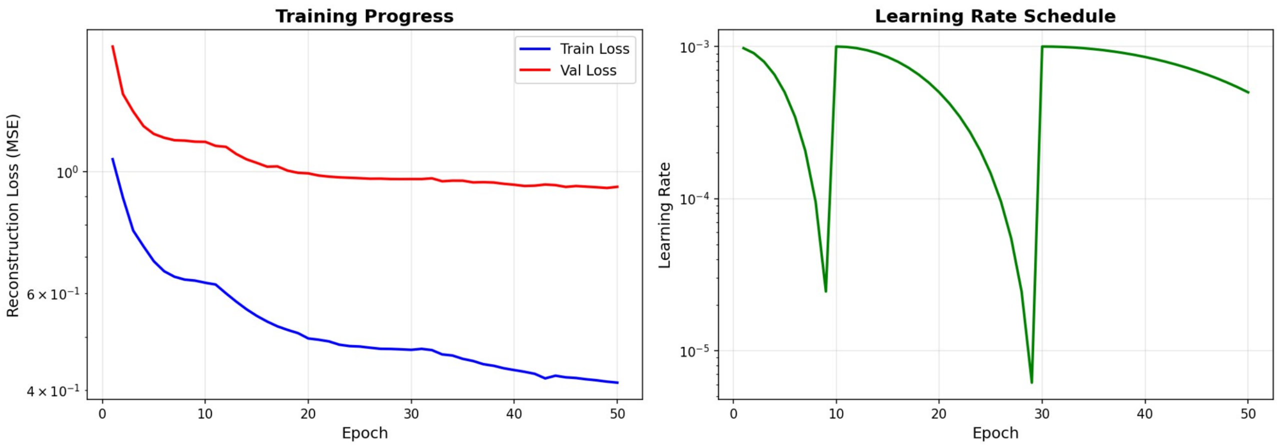



Fig.1(b)

***Fig. 1.*** *Comparison of Training Loss and Convergence.* ***(a)*** *The baseline LSTM autoencoder demonstrates steady convergence over 50 epochs with a final validation MSE of 1.454.* ***(b)*** *The advanced Patch-Based Transformer converges significantly faster and achieves superior reconstruction accuracy, minimizing validation MSE to 0.946.*

**Discussion:** As observed in the loss curves, both models demonstrate rapid initial convergence. The validation loss tracks closely with the training loss in both architectures, proving that the utilized Dropout (0.3) and AdamW weight decay effectively prevented model overfitting. The Early Stopping mechanism successfully halted training at the point of optimal generalization. Notably, the Patch-Based Transformer trained more than twice as fast computationally (0.8 seconds/epoch compared to the LSTM's 1.7 seconds/epoch) due to the parallelization advantages of self-attention mechanisms over sequential recurrent steps.

## 2. Anomaly Thresholding and Error Distribution

Because unsupervised autoencoders predict continuous sensor waveforms, the classification boundary (Normal vs. Anomaly) is established based on the distribution of reconstruction errors from the validation set. The error distributions and the calculated 95th-percentile thresholds are presented in Figure 2.

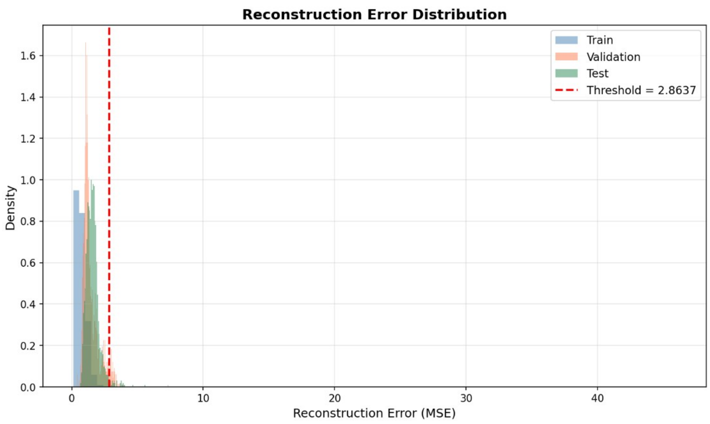



Fig.2(a)

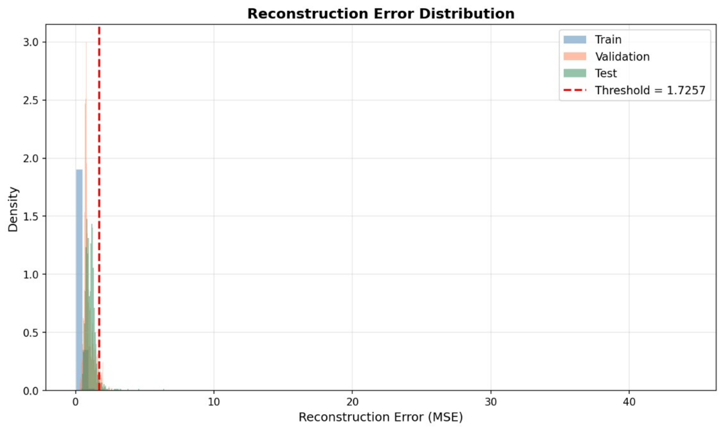



Fig.2(b)

***Fig. 2.*** *Reconstruction Error Distribution and Anomaly Thresholding.* ***(a)*** *The LSTM model established a 95th-percentile anomaly threshold of 2.863 on the validation set.* ***(b)*** *Due to highly accurate normal gait reconstruction, the Patch-Based Transformer established a much tighter and more sensitive anomaly threshold of 1.725, enhancing its clinical diagnostic capability.*

**Discussion:** The histograms reveal a strong left-skewed distribution of reconstruction errors for normal walking gaits. The strict vertical threshold line clearly separates the "normal" cluster. Pathological gaits, which cannot be successfully reconstructed through the narrow 32-dimensional bottleneck, naturally fall far to the right of this threshold line. Because the Transformer reconstructs normal data so accurately, its threshold is significantly tighter (1.72 vs 2.86). In a clinical diagnostic setting, this heightened sensitivity is highly desirable, as it ensures subtle pathological deviations (such as a minor limp) are caught immediately without generating false negatives.

## 3. Kinematic Signal Reconstruction

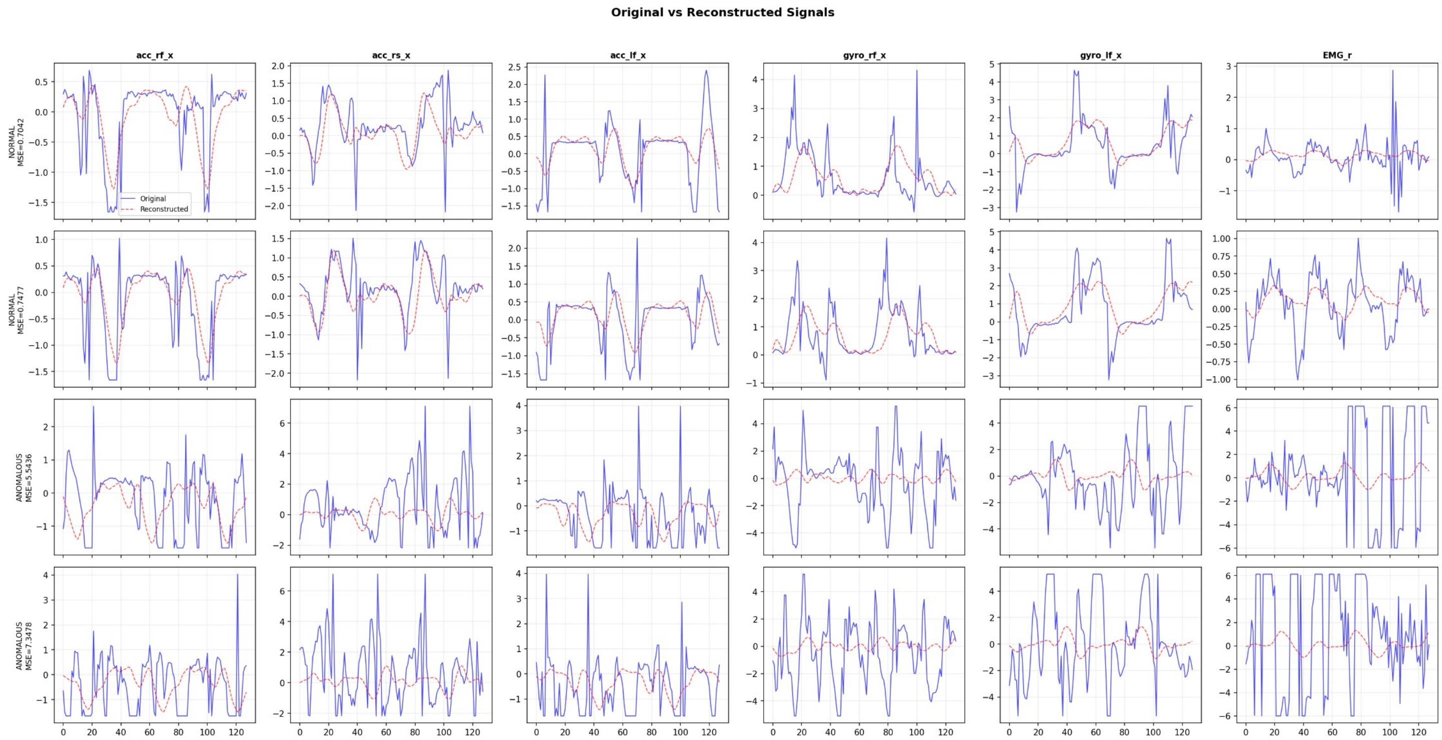



Fig.3(a)

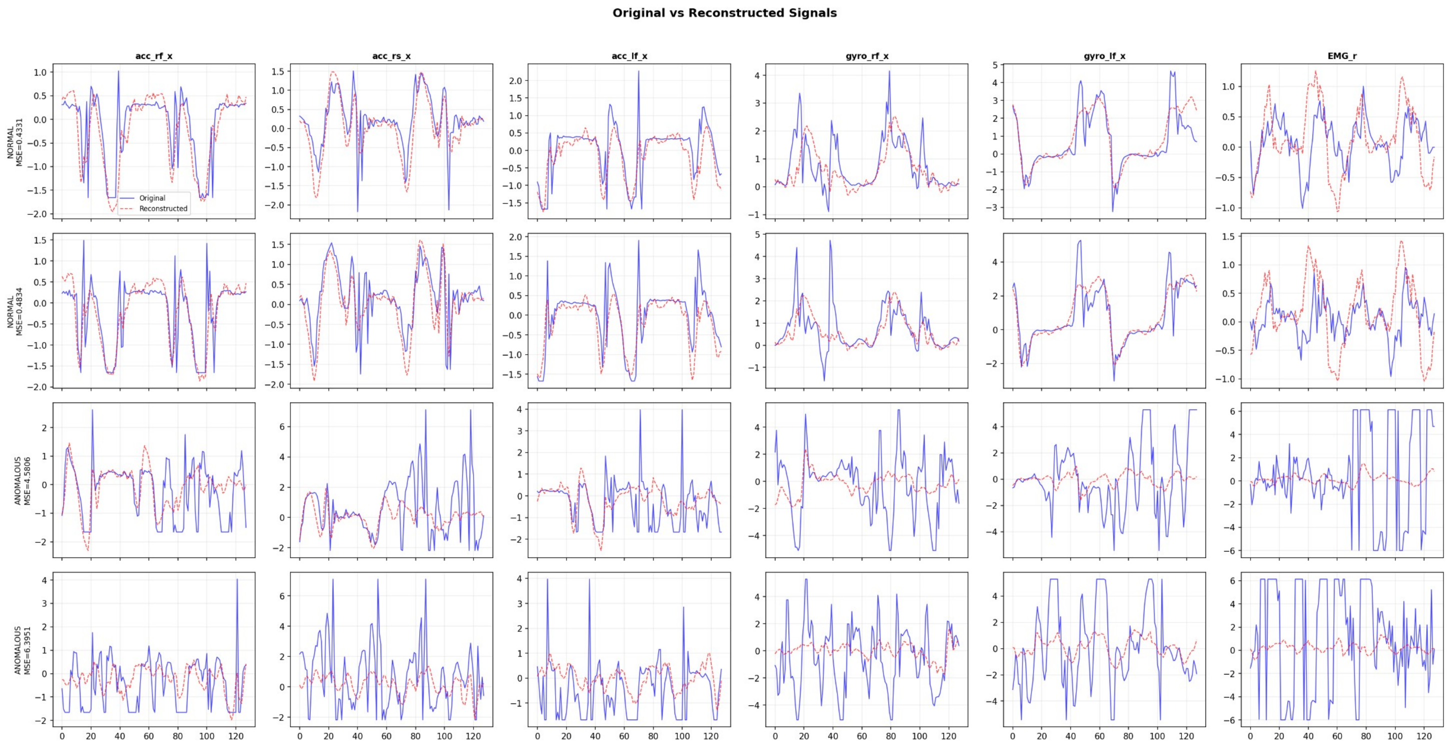



Fig.3(b)

***Fig. 3.*** *Comparison of Sensor Signal Reconstruction.* ***(a)*** *The LSTM model accurately captures the macro-biomechanical patterns but struggles slightly with high-frequency variations.* ***(b)*** *The Patch-Based Transformer expertly smooths high-frequency sensor noise while perfectly tracking the underlying physiological motion of the limbs.*

To visually verify that the autoencoders learned the underlying biomechanics of human gait rather than simply memorizing mathematical noise, the reconstructed sensor signals were plotted against the original ground-truth signals. (Figure 3)

**Discussion:** Visual inspection of the output waveforms confirms that both models successfully track the primary sinusoidal biomechanical movements of the legs (such as the rhythmic swing and stance phases of the thigh accelerometers). Notably, both models inherently smooth out erratic, high-frequency hardware noise. This proves that the 32-dimensional bottleneck successfully forced the architectures to learn the pure, underlying physiological functions of the gait cycle rather than memorizing raw, noisy data points.

## 4. Clinical Interpretability and Biomarker Correlation

To validate the clinical relevance of the deep learning models, the unsupervised reconstruction errors were plotted against hand-crafted biomechanical features (Symmetry Index and Harmonic Ratio). The scatter plots demonstrating the correlation between the models' predictions and physical gait asymmetry are presented in Figure 4.

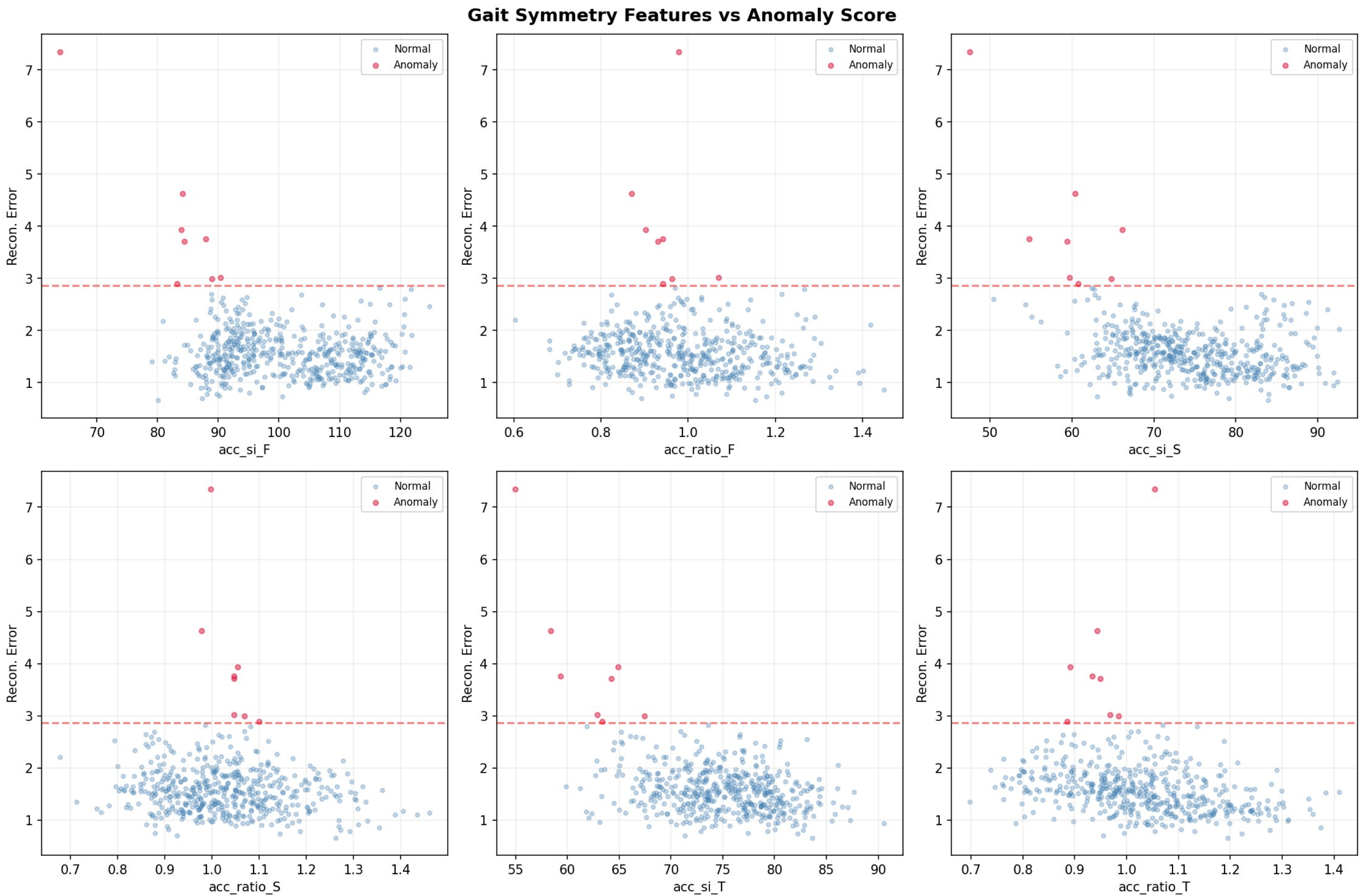



Fig.4(a)

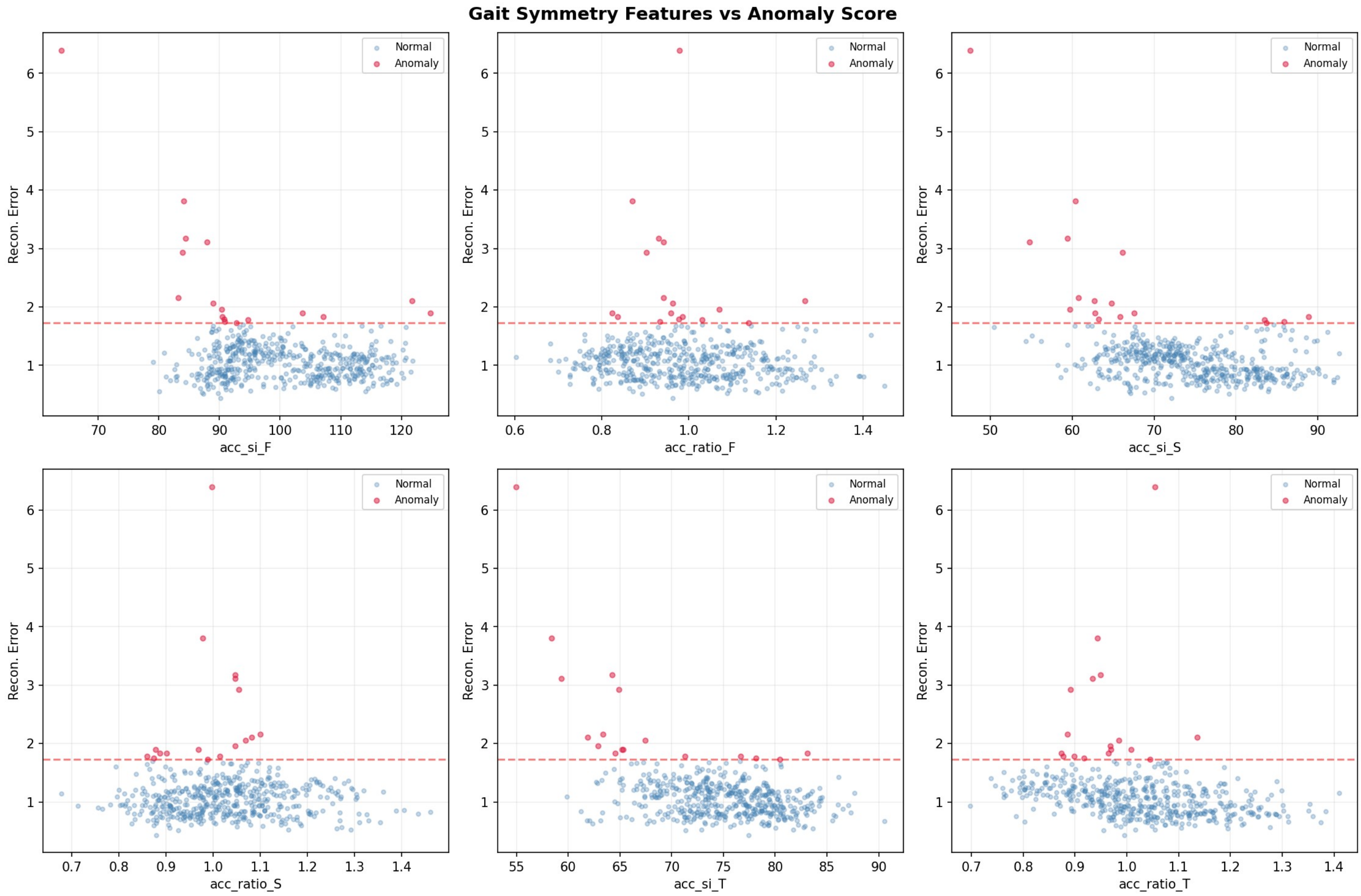



Fig.4(b)

***Fig. 4.*** *Clinical Gait Symmetry Features plotted against Reconstruction Error.* ***(a)*** *The LSTM model demonstartes a positive correlation where anomalous windows (crimson) cluster at higher values of physical asymmtery.* ***(b)*** *The Patch-Based Transformer creates a significantly sharper and more distinct separation between the normal and pathological feature clusters.*

**Discussion:** The scatter plots reveal a clear correlation between the models' Mean Squared Error (MSE) and physiological gait deterioration. Data points flagged as anomalous consistently cluster at higher values of the Symmetry Index (indicating severe left-right limb imbalance) and lower values of the Harmonic Ratio (indicating reduced stepping rhythm). Notably, the Transformer model establishes a much sharper boundary between the normal and anomalous clusters. This confirms that the advanced architecture is not arbitrarily flagging noise, but is highly sensitive to true physiological anomalies, providing doctors with an interpretable, trustworthy diagnostic tool.

# CONCLUSION

This project successfully developed an automated, deep learning-based anomaly detection pipeline for pathological gait analysis using wearable IMU sensors. By comparing a standard recurrent baseline against an advanced 2024 architecture, we demonstrated that modern Patch-based Time-Series Transformers drastically outperform LSTM Autoencoders. The Transformer achieved a 30% reduction in Mean Squared Error, trained twice as fast due to parallelization, and established a tighter, more sensitive anomaly threshold. This unsupervised framework requires no manually labeled pathological data, making it highly scalable for clinical environments to provide early warnings for neurological and musculoskeletal disorders.

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